Part 6

Getting Started with GNU Octave

In the first three parts of this series, we started out learning Octave by looking at some of the major functions it provides. In the fourth part, we explored scripting in Octave, where we used the knowledge acquired in the previous articles to build larger Octave programs. Last month, we took a look at how we could interface C++ and Octave. In this final part of the series, we briefly cover Octave packages, and the compatibility aspects between Octave and MATLAB.

Octave packages enable Octave users to share their Octave code with the user community. Octave-Forge (*http://octave.sourceforge.net/*) is where all community-developed packages are submitted and then made available to the Octave community. Octave packages supplement the core functionality provided by Octave. The list of currently available packages can be seen at *http://octave.sourceforge.net/packages.php*

pkg is the Octave command used to work with Octave packages. To find more information on it, type help *pkg* at the Octave command line.

Installation

As an exercise, we shall install the *sockets* package from *http://octave.sourceforge.net/sockets/*. This is an interesting package, which adds support for socket programming in Octave. Download the package, which is in the *.tar.gz* file format. Octave needs the package file in the current working directory to be able to install it. While at the Octave command-line, you can check your current working directory with *pwd*. If it is the same directory as the one in which you saved the package file, then go ahead with the install—otherwise, change the Octave current working directory with the *cd* command.

The following is a transcript of an attempt to install the package, at the Octave command-line:

```
octave-3.2.3:15> pkg install sockets-1.0.6.tar.gz
couldn't create installation directory /usr/lib/octave/
packages/3.2/sockets-1.0.6 : Permission denied
```